DevOps and Cloud Assignment

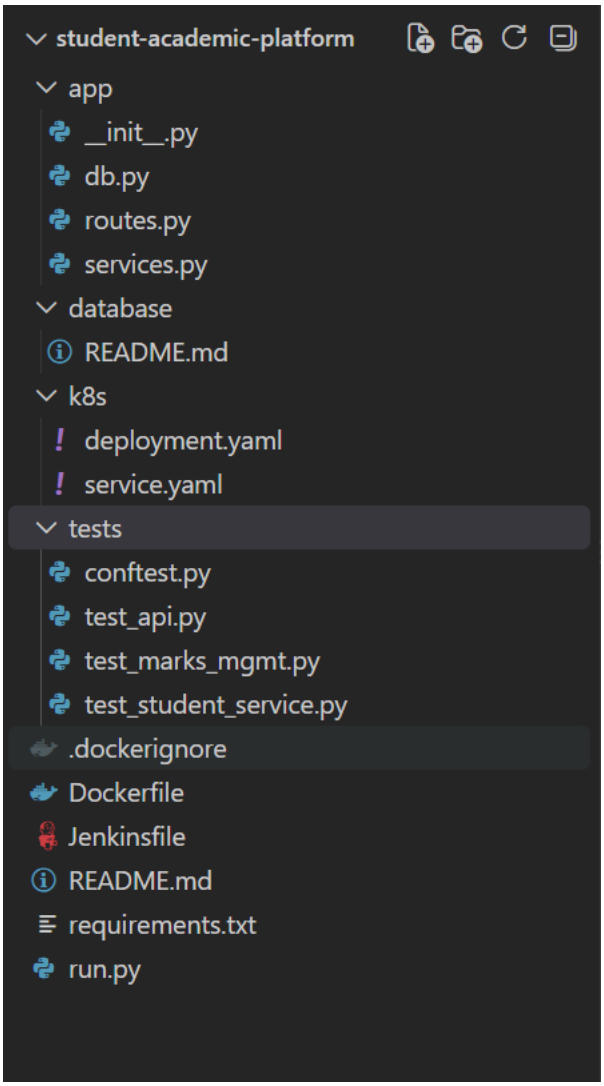
"Student Academic Management Platform using DevOps Automation"

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Folder Structure



Requirements.txt

requirements.txt

```
Flask==3.1.1
pytest==8.4.1
pytest-cov==6.2.1
ruff==0.12.3

```

init.py

```python
import os
from flask import Flask
from .routes import api
from .db import init_db

def create_app(test_config=None):
    app = Flask(__name__)
    app.config.update(
        DATABASE=os.getenv("DATABASE", "student_platform.db"),
        TESTING=False,
    )
    if test_config:
        app.config.update(test_config)
    init_db(app)
    app.register_blueprint(api, url_prefix="/api")
    return app

```

db.py

db.py

```python
import sqlite3
from flask import current_app, g

SCHEMA = """
CREATE TABLE IF NOT EXISTS students (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT NOT NULL,
    email TEXT NOT NULL UNIQUE
);
CREATE TABLE IF NOT EXISTS faculty (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT NOT NULL,
    department TEXT NOT NULL
);
CREATE TABLE IF NOT EXISTS courses (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    code TEXT NOT NULL UNIQUE,
    title TEXT NOT NULL,
    credits INTEGER NOT NULL CHECK (credits > 0)
);
CREATE TABLE IF NOT EXISTS enrollments (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    student_id INTEGER NOT NULL REFERENCES students(id),
    course_id INTEGER NOT NULL REFERENCES courses(id),
    semester TEXT NOT NULL,
```

routes.py

```python
from flask import Blueprint, jsonify, request
from . import services

api = Blueprint("api", __name__)

def error(message, status=400):
    return jsonify({"error": message}), status

@api.get("/health")
def health():
    return jsonify({"status": "healthy", "service": "student-academic-platform"}), 200

@api.get("/")
def index():
    return jsonify({
        "application": "Student Academic Management Platform",
        "endpoints": [
            "GET /api/health", "GET/POST /api/students", "GET/POST /api/faculty",
            "GET/POST /api/courses", "POST /api/enrollments", "POST /api/marks",
            "GET /api/students/<id>/gpa?semester=..."
        ]
    })
```

services.py

services.py

app > services.py > ...

```python
from .db import get_db

MAX_CREDITS_PER_SEMESTER = 18

def add_student(name, email):
    name, email = name.strip(), email.strip().lower()
    if not name or not email or "@" not in email:
        raise ValueError("A valid student name and email are required.")
    db = get_db()
    try:
        cur = db.execute("INSERT INTO students(name, email) VALUES (?, ?)", (name, emai
        db.commit()
        return cur.lastrowid
    except Exception as exc:
        db.rollback()
        if "UNIQUE constraint failed" in str(exc):
            raise ValueError("A student with this email already exists.") from exc
        raise

def add_faculty(name, department):
    if not name.strip() or not department.strip():
        raise ValueError("Faculty name and department are required.")
    db = get_db()
    cur = db.execute("INSERT INTO faculty(name, department) VALUES (?, ?)",
                     (name.strip(), department.strip()))
```

Run.py

run.py

run.py > ...

```python
from app import create_app

app = create_app()

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)

```

Dockerfile

Dockerfile

```
FROM python:3.12-slim
ENV PYTHONDONTWRITEBYTECODE=1
ENV PYTHONUNBUFFERED=1
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app ./app
COPY run.py .
RUN mkdir -p /app/database
ENV DATABASE=/app/database/student_platform.db
EXPOSE 5000
CMD ["python", "run.py"]

```

Jenkinsfile

```
pipeline {
    agent any

    stages {
        stage('1. Checkout Code') {
            steps {
                echo 'Checking out Student Academic Management Platform source code'
                checkout scm
            }
        }

        stage('2. Build') {
            steps {
                sh 'python3 --version'
                sh 'python3 -m venv .venv'
                sh '. .venv/bin/activate && pip install --upgrade pip'
                sh '. .venv/bin/activate && pip install -r requirements.txt'
                sh '. .venv/bin/activate && ruff check app tests run.py --output-format concise
            }
        }

        stage('3. Unit Test') {
            steps {
                sh '. .venv/bin/activate && pytest -v --junitxml=pytest-report.xml --co=app --
            }
```

Deployment.yaml

deployment.yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: student-academic-deployment
  labels:
    app: student-academic-platform
spec:
  replicas: 3
  selector:
    matchLabels:
      app: student-academic-platform
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1
      maxUnavailable: 0
  template:
    metadata:
      labels:
        app: student-academic-platform
    spec:
      containers:
        - name: student-academic-app
          image: student-academic-platform:v1.0.0
```

Service.yaml

service.yaml

```yaml
apiVersion: v1
kind: Service
metadata:
  name: student-academic-service
  labels:
    app: student-academic-platform
spec:
  type: NodePort
  selector:
    app: student-academic-platform
  ports:
    - protocol: TCP
      port: 5000
      targetPort: 5000
      nodePort: 30090
```

Conftest.py test_api.*py* test_marks_mgmt.py test_student_service.py

```
anushka@DESKTOP-C5NI3F1:/mnt/c/Users/DELL/Downloads/student-academic-platform$ cat tests/conftest.py
cat tests/test_api.py
cat tests/test_marks_mgmt.py
cat tests/test_student_service.py
import pytest
from app import create_app

@pytest.fixture()
def app(tmp_path):
    db_file = tmp_path / "test.db"
    application = create_app({"TESTING": True, "DATABASE": str(db_file)})
    yield application

@pytest.fixture()
def client(app):
    return app.test_client()
def test_health_endpoint(client):
    response = client.get("/api/health")
    assert response.status_code == 200
    assert response.get_json()["status"] == "healthy"

def test_create_student_endpoint(client):
    response = client.post("/api/students", json={
        "name": "Asha Patil", "email": "asha@example.com"
    })
```

Create an isolated Python environment

```
root@DESKTOP-C5NI3F1:/mnt/c/Users/DELL/Downloads/student-academic-platform# python3 -m venv .venv
source .venv/bin/activate
(.venv) root@DESKTOP-C5NI3F1:/mnt/c/Users/DELL/Downloads/student-academic-platform#
```

Install the required dependencies

```
(.venv) root@DESKTOP-C5NI3F1:/mnt/c/Users/DELL/Downloads/student-academic-platform# python -m pip install --upgrade pip
pip install -r requirements.txt
Requirement already satisfied: pip in ./.venv/lib/python3.14/site-packages (25.1.1)
Collecting pip
  Downloading pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2.1-py3-none-any.whl (1.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 5.6 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.1.1
    Uninstalling pip-25.1.1:
      Successfully uninstalled pip-25.1.1
Successfully installed pip-26.2.1
Collecting Flask==3.1.1 (from -r requirements.txt (line 1))
  Downloading flask-3.1.1-py3-none-any.whl.metadata (3.0 kB)
Collecting pytest==8.4.1 (from -r requirements.txt (line 2))
  Downloading pytest-8.4.1-py3-none-any.whl.metadata (7.7 kB)
Collecting pytest-cov==6.2.1 (from -r requirements.txt (line 3))
  Downloading pytest_cov-6.2.1-py3-none-any.whl.metadata (30 kB)
```

Verify existing automated tests

```
(.venv) root@DESKTOP-C5NI3F1:/mnt/c/Users/DELL/Downloads/student-academic-platform# python -m pytest -v
============================================================ test session starts ============================================================
platform linux -- Python 3.14.4, pytest-8.4.1, pluggy-1.6.0 -- /mnt/c/Users/DELL/Downloads/student-academic-platform/.venv/bin/python
cachedir: .pytest_cache
rootdir: /mnt/c/Users/DELL/Downloads/student-academic-platform
plugins: cov-6.2.1
collected 10 items

tests/test_api.py::test_health_endpoint PASSED                                                                                      [ 10%]
tests/test_api.py::test_create_student_endpoint PASSED                                                                              [ 20%]
tests/test_api.py::test_invalid_student_is_rejected PASSED                                                                          [ 30%]
tests/test_marks_mgmt.py::test_grade_point_bands PASSED                                                                             [ 40%]
tests/test_marks_mgmt.py::test_marks_and_weighted_gpa PASSED                                                                        [ 50%]
tests/test_marks_mgmt.py::test_marks_require_enrollment PASSED                                                                      [ 60%]
tests/test_student_service.py::test_student_can_be_created PASSED                                                                   [ 70%]
tests/test_student_service.py::test_duplicate_student_email_is_rejected PASSED                                                      [ 80%]
tests/test_student_service.py::test_enrollment_cannot_exceed_18_credits PASSED                                                      [ 90%]
tests/test_student_service.py::test_same_course_cannot_be_enrolled_twice PASSED                                                     [100%]

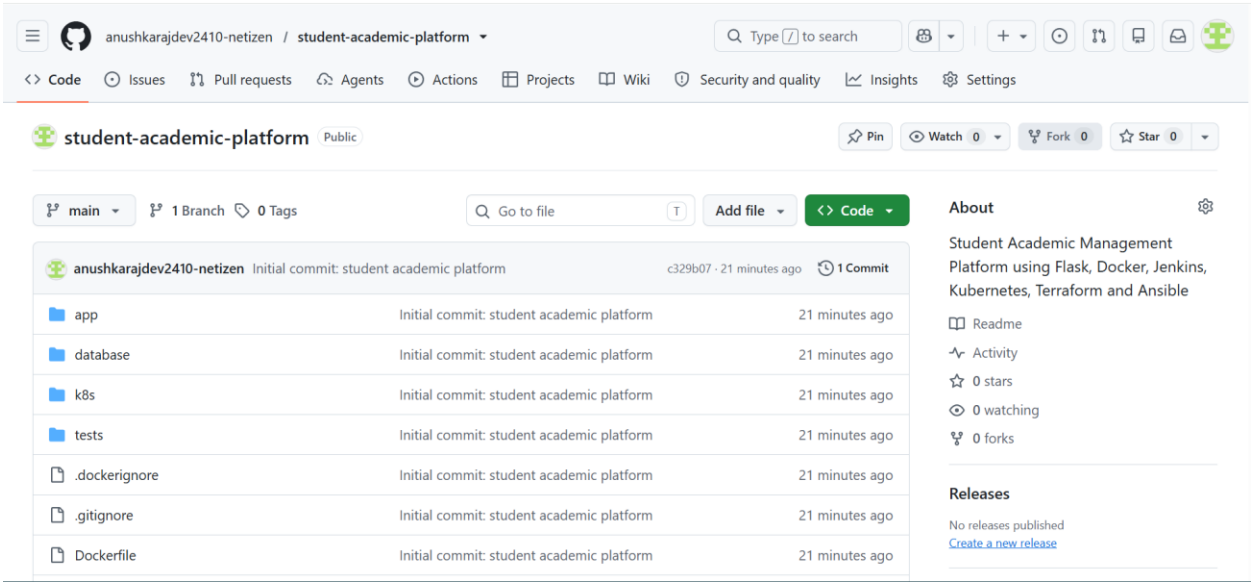
============================================================ 10 passed in 0.70s =============================================================
(.venv) root@DESKTOP-C5NI3F1:/mnt/c/Users/DELL/Downloads/student-academic-platform#
```

Configure your Git identity

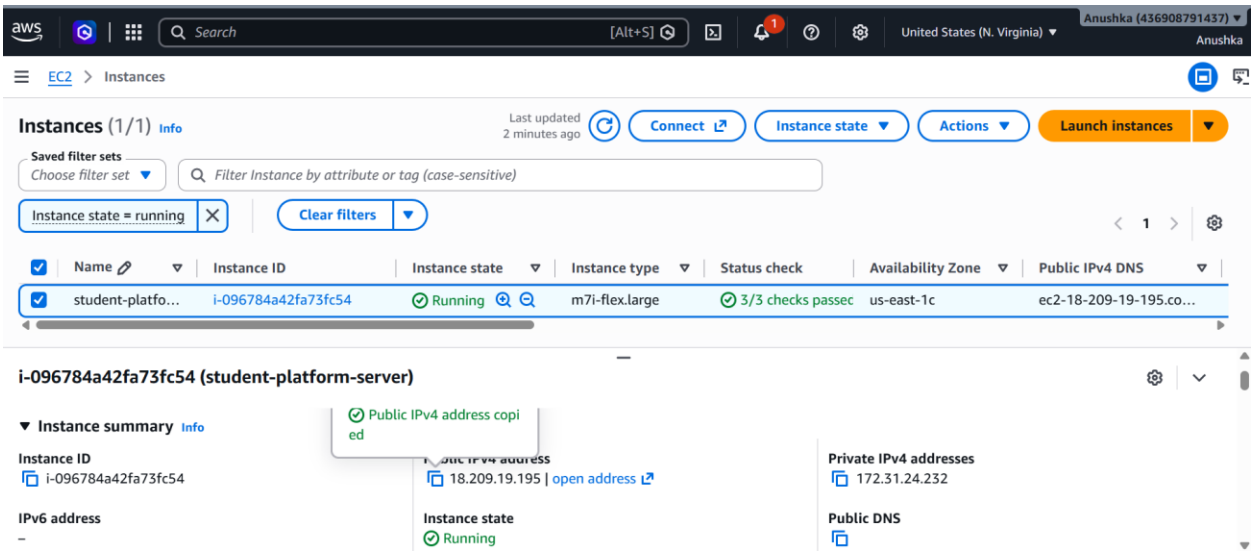
```
(.venv) root@DESKTOP-C5NI3F1:/mnt/c/Users/DELL/Downloads/student-academic-platform# git config --global us
er.name "Anushka Rajdev"
git config --global user.email "anushkarajdev2410@gmail.com"
(.venv) root@DESKTOP-C5NI3F1:/mnt/c/Users/DELL/Downloads/student-academic-platform# git config --global --
list
user.name=Anushka Rajdev
user.email=anushkarajdev2410@gmail.com
(.venv) root@DESKTOP-C5NI3F1:/mnt/c/Users/DELL/Downloads/student-academic-platform#
```

Create and Setup a Github Repository

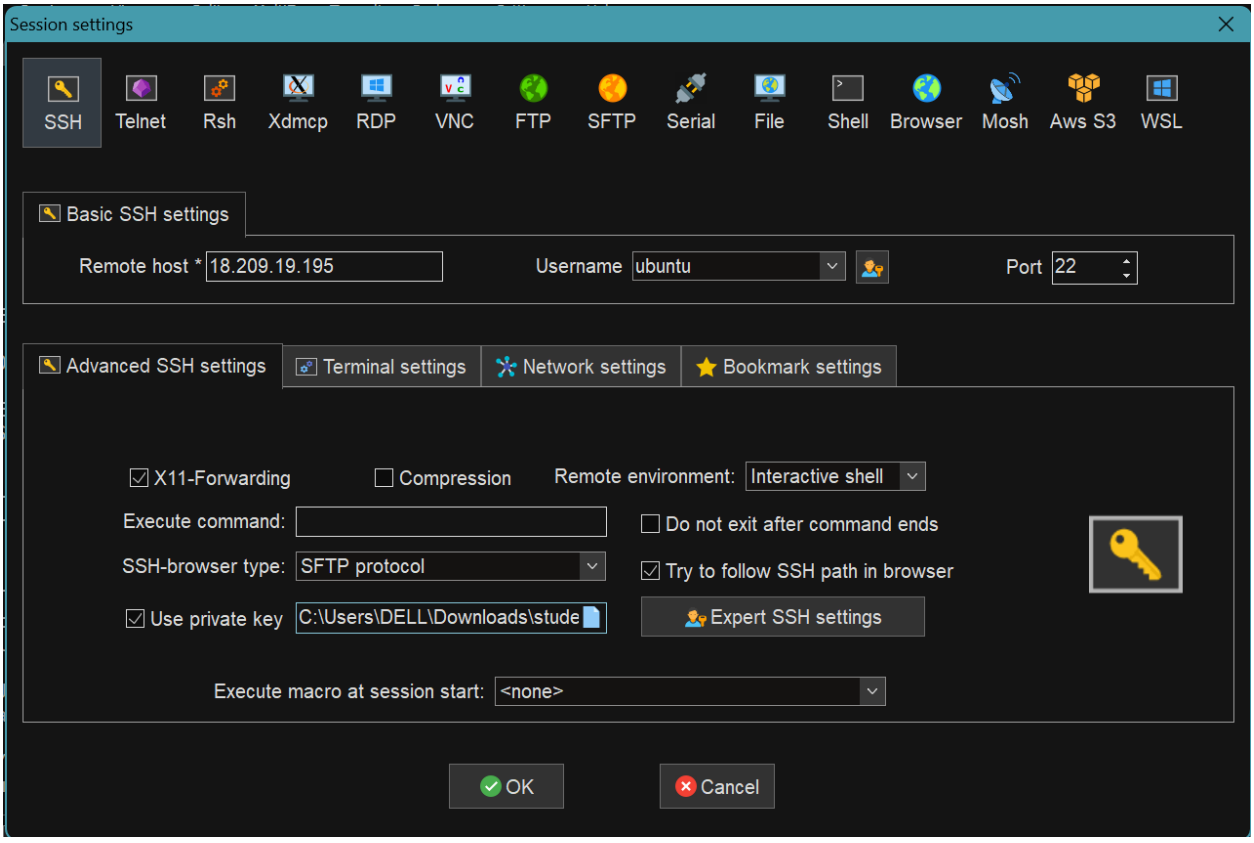
https://github.com/anushkarajdev2410-netizen/student-academic-platform.git



Create and configure an EC2 Instance on AWS



Connect to Mobaxterm



sudo apt install -y docker.io

```
ubuntu@ip-172-31-24-232:~$ sudo apt update
sudo apt upgrade -y
sudo apt install -y git python3 python3-pip python3-venv python3-dev build-essential curl wget ca-certificates gnupg
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease [137 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 Packages [16.3 MB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64v3 Packages [602 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [138 kB]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [4616 B]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64v3 Packages [201 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [65.4 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [55.5 kB]
```

Install Docker

```
ubuntu@ip-172-31-24-232:~$ sudo apt install -y docker.io
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Installing:
  docker.io

Installing dependencies:
  bridge-utils  containerd  dns-root-data  dnsmasq-base  pigz  runc  ubuntu-fan

Suggested packages:
  ifupdown    cgroupfs-mount  debootstrap    docker-compose-v2  rinse        zfs-fuse
  aufs-tools  | cgroup-lite   docker-buildx  docker-doc         rootlesskit  | zfsutils

Summary:
  Upgrading: 0, Installing: 8, Removing: 0, Not Upgrading: 2
  Download size: 74.0 MB
  Space needed: 281 MB / 26.8 GB available
```

Verify Docker and Enable Docker

```
ubuntu@ip-172-31-24-232:~$ sudo systemctl enable --now docker
ubuntu@ip-172-31-24-232:~$ sudo usermod -aG docker ubuntu
ubuntu@ip-172-31-24-232:~$ newgrp docker
ubuntu@ip-172-31-24-232:~$ docker --version
docker ps
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-24-232:~$
```

Download the Project from Github:

```
ubuntu@ip-172-31-24-232:~$ cd ~
git clone https://github.com/anushkarajdev2410-netizen/student-academic-platform.git
Cloning into 'student-academic-platform'...
remote: Enumerating objects: 24, done.
remote: Counting objects: 100% (24/24), done.
remote: Compressing objects: 100% (23/23), done.
remote: Total 24 (delta 0), reused 24 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (24/24), 7.90 KiB | 1.13 MiB/s, done.
ubuntu@ip-172-31-24-232:~$
```

```
ubuntu@ip-172-31-24-232:~/student-academic-platform$ ls -la
total 56
drwxrwxr-x 7 ubuntu docker 4096 Oct 12 15:25 .
drwxr-x--- 5 ubuntu ubuntu 4096 Oct 12 15:25 ..
-rw-rw-r-- 1 ubuntu docker   75 Oct 12 15:25 .dockerignore
drwxrwxr-x 7 ubuntu docker 4096 Oct 12 15:25 .git
-rw-rw-r-- 1 ubuntu docker  275 Oct 12 15:25 .gitignore
-rw-rw-r-- 1 ubuntu docker  303 Oct 12 15:25 Dockerfile
-rw-rw-r-- 1 ubuntu docker 1832 Oct 12 15:25 Jenkinsfile
-rw-rw-r-- 1 ubuntu docker 1164 Oct 12 15:25 README.md
drwxrwxr-x 2 ubuntu docker 4096 Oct 12 15:25 app
drwxrwxr-x 2 ubuntu docker 4096 Oct 12 15:25 database
drwxrwxr-x 2 ubuntu docker 4096 Oct 12 15:25 k8s
-rw-rw-r-- 1 ubuntu docker   58 Oct 12 15:25 requirements.txt
-rw-rw-r-- 1 ubuntu docker  114 Oct 12 15:25 run.py
drwxrwxr-x 2 ubuntu docker 4096 Oct 12 15:25 tests
ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Create an isolated Python environment and Install the Python dependencies

```
ubuntu@ip-172-31-24-232:~/student-academic-platform$ python3 -m venv .venv
source .venv/bin/activate
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ python -m pip install --upgrade pip
Requirement already satisfied: pip in ./.venv/lib/python3.14/site-packages (25.1.1)
Collecting pip
  Downloading pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2.1-py3-none-any.whl (1.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 57.0 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 25.1.1
    Uninstalling pip-25.1.1:
      Successfully uninstalled pip-25.1.1
Successfully installed pip-26.2.1
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ pip install -r requirements.txt
pip install pytest
Collecting Flask==3.1.1 (from -r requirements.txt (line 1))
  Downloading flask-3.1.1-py3-none-any.whl.metadata (3.0 kB)
Collecting pytest==8.4.1 (from -r requirements.txt (line 2))
  Downloading pytest-8.4.1-py3-none-any.whl.metadata (7.7 kB)
Collecting pytest-cov==6.2.1 (from -r requirements.txt (line 3))
```

Run the project tests

```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ python -m pytest -v
============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-8.4.1, pluggy-1.6.0 -- /home/ubuntu/student-academic-platform/.venv/bin/python
cachedir: .pytest_cache
rootdir: /home/ubuntu/student-academic-platform
plugins: cov-6.2.1
collected 10 items

tests/test_api.py::test_health_endpoint PASSED                           [ 10%]
tests/test_api.py::test_create_student_endpoint PASSED                   [ 20%]
tests/test_api.py::test_invalid_student_is_rejected PASSED               [ 30%]
tests/test_marks_mgmt.py::test_grade_point_bands PASSED                  [ 40%]
tests/test_marks_mgmt.py::test_marks_and_weighted_gpa PASSED             [ 50%]
tests/test_marks_mgmt.py::test_marks_require_enrollment PASSED           [ 60%]
tests/test_student_service.py::test_student_can_be_created PASSED        [ 70%]
tests/test_student_service.py::test_duplicate_student_email_is_rejected PASSED [ 80%]
tests/test_student_service.py::test_enrollment_cannot_exceed_18_credits PASSED [ 90%]
tests/test_student_service.py::test_same_course_cannot_be_enrolled_twice PASSED [100%]

============================== 10 passed in 0.07s ==============================
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Build Docker image

```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ docker build -t student-academic-platform:v1.0.0 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  83.97kB
Step 1/12 : FROM python:3.12-slim
 ---> a6e34c598f24
Step 2/12 : ENV PYTHONDONTWRITEBYTECODE=1
 ---> Using cache
 ---> 44e065272104
Step 3/12 : ENV PYTHONUNBUFFERED=1
 ---> Using cache
 ---> c9a23088d154
Step 4/12 : WORKDIR /app
```

Verify Docker Image

```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ docker images

IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
python:3.12-slim                     a6e34c598f24        179MB         45.4MB
student-academic-platform:v1.0.0     a13f0dd636c6        250MB         61.6MB
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Start the application container

```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ docker run -d \
  --name student-academic-platform \
  --restart unless-stopped \
  -p 5000:5000 \
  student-academic-platform:v1.0.0
7d0305904eb269e493675c5ad464a9949926c6c8cd02fa3035699b8635eaf3cd
```

Make application publicly accessible

Install Nginx

```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo apt update
sudo apt install -y nginx
sudo systemctl enable --now nginx
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap  nginx-doc  ssl-cert
```

Check Status

```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo systemctl status nginx --no-pager
● nginx.service - A high performance web server and a reverse proxy server
     Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-10-12 15:46:34 UTC; 51s ago
 Invocation: 48577ac5d3714dcf8bc78a9a1b1f35c7
       Docs: man:nginx(8)
   Main PID: 42460 (nginx)
      Tasks: 3 (limit: 8157)
     Memory: 3M (peak: 7.3M)
        CPU: 29ms
     CGroup: /system.slice/nginx.service
             ├─42460 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             ├─42462 "nginx: worker process"
             └─42463 "nginx: worker process"

Oct 12 15:46:34 ip-172-31-24-232 systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy server...
Oct 12 15:46:34 ip-172-31-24-232 systemd[1]: Started nginx.service - A high performance web server and a reverse proxy server.
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Configure Nginx to forward requests to Flask

```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo nano /etc/nginx/sites-available/student-academic-platform
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo ln -s /etc/nginx/sites-available/student-academic-platform /etc/n
ginx/sites-enabled/student-academic-platform
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo rm -f /etc/nginx/sites-enabled/default
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Test and reload Nginx:

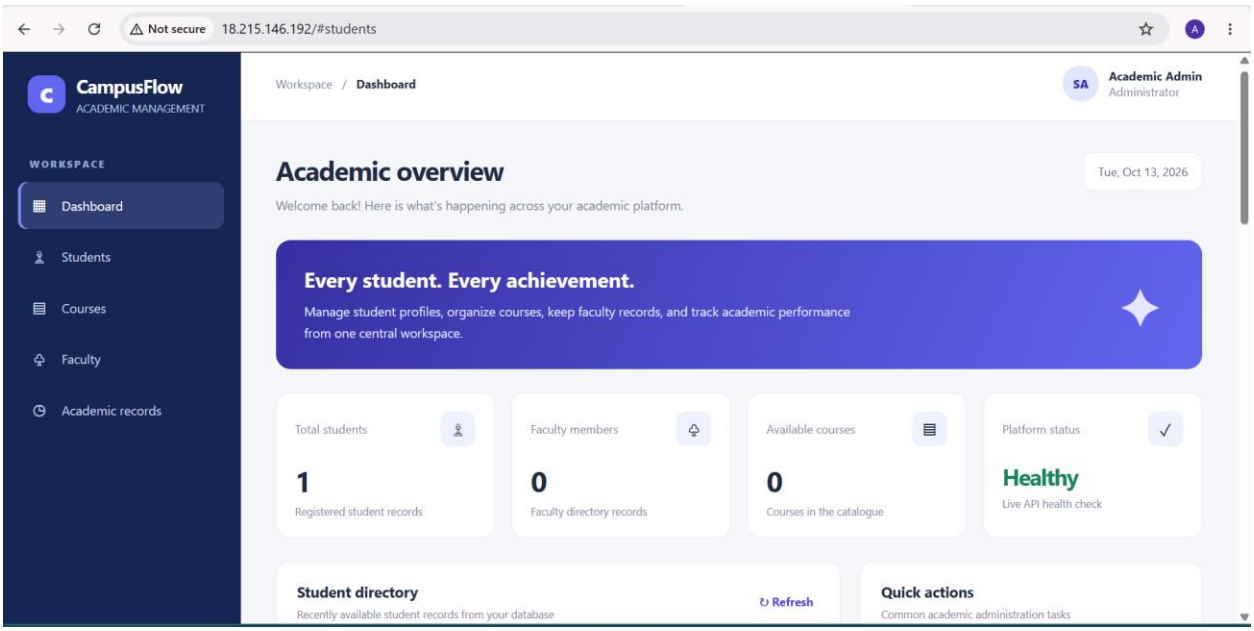
```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo nginx -t
sudo systemctl reload nginx
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

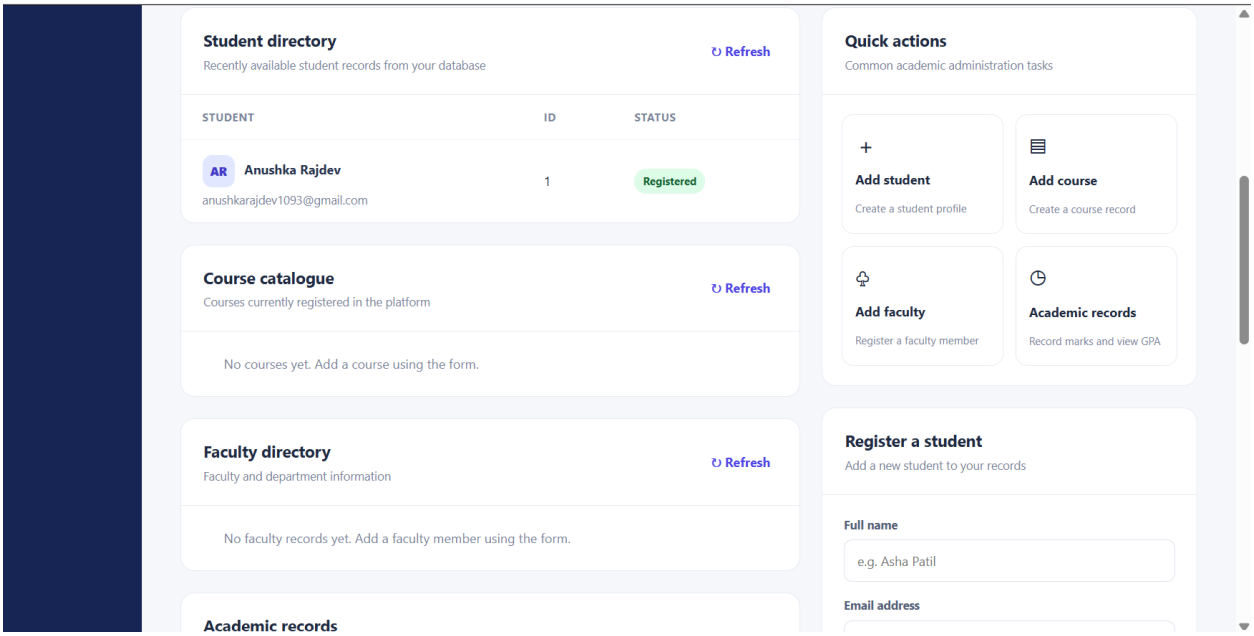
Test the public website:

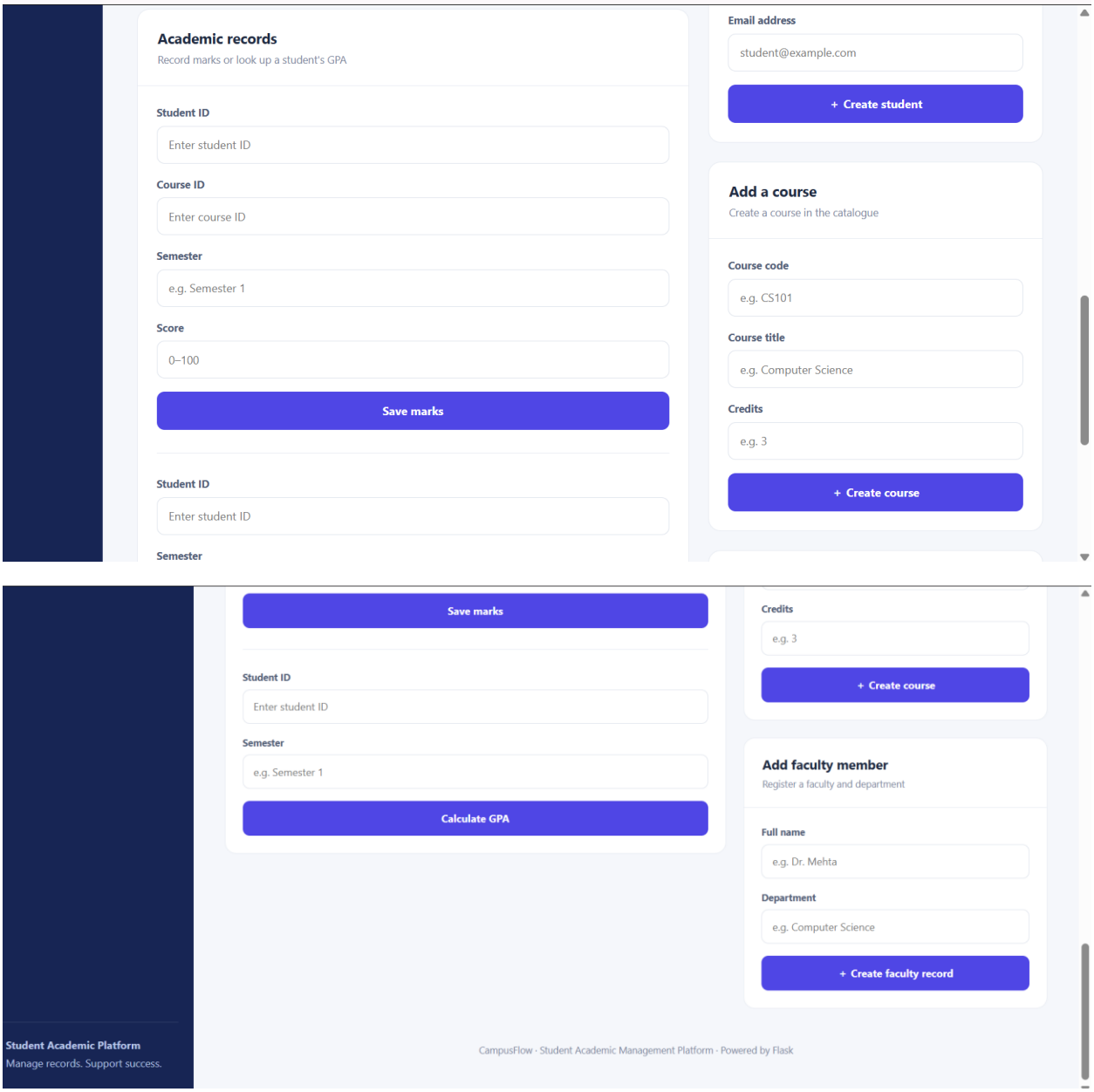
```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ curl -I http://localhost:5000/
curl -I http://localhost/
HTTP/1.1 200 OK
Server: Werkzeug/3.1.9 Python/3.12.15
Date: Mon, 12 Oct 2026 18:36:46 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 25295
Connection: close

HTTP/1.1 200 OK
Server: nginx/1.28.3 (Ubuntu)
Date: Mon, 12 Oct 2026 18:36:46 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 25295
Connection: keep-alive
```

Check the Website on browser:


Not secure
18.215.146.192/#students
CampusFlow
ACADEMIC MANAGEMENT
WORKSPACE
Dashboard
Students
Courses
Faculty
Academic records
Workspace / Dashboard
SA
Academic Admin
Administrator
Academic overview
Tue, Oct 13, 2026
Welcome back! Here is what's happening across your academic platform.
Every student. Every achievement.
Manage student profiles, organize courses, keep faculty records, and track academic performance from one central workspace.
Total students
1
Registered student records
Faculty members
0
Faculty directory records
Available courses
0
Courses in the catalogue
Platform status
Healthy
Live API health check
Student directory
Recently available student records from your database
Refresh
Quick actions
Common academic administration tasks


Student directory
Recently available student records from your database
Refresh
STUDENT
ID
STATUS
AR
Anushka Rajdev
anushkarajdev1093@gmail.com
1
Registered
Course catalogue
Courses currently registered in the platform
Refresh
No courses yet. Add a course using the form.
Faculty directory
Faculty and department information
Refresh
No faculty records yet. Add a faculty member using the form.
Academic records
Quick actions
Common academic administration tasks
Add student
Create a student profile
Add course
Create a course record
Add faculty
Register a faculty member
Academic records
Record marks and view GPA
Register a student
Add a new student to your records
Full name
e.g. Asha Patil
Email address



Install Java 21:

```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo apt update
sudo apt install -y fontconfig openjdk-21-jre
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Installing:
  fontconfig  openjdk-21-jre

Installing dependencies:
  adwaita-icon-theme        libatk-wrapper-java      libglx-mesa0          librsvg2-2          libxfixes3
  alsa-topology-conf        libatk-wrapper-java-jni  libglx0               librsvg2-common     libxft2
  alsa-ucm-conf             libatk1.0-0t64           libglycin-2-0         libsharpyuv0        libxi6
  at-spi2-common            libatspi2.0-0t64         libgraphite2-3        libsm6              libxinerama1
  at-spi2-core              libavahi-client3         libgtk-3-0t64         libthai-data        libxkbfile1
  bubblewrap                libavahi-common-data     libgtk-3-bin          libthai0            libxmu6
  ca-certificates-java      libavahi-common3         libgtk-3-common       libvulkan1          libxpm4
  dconf-gsettings-backend   libcairo-gobject2        libharfbuzz0b         libwayland-client0  libxrandr2
  dconf-service             libcairo2                libheif-plugin-aomdec libwayland-cursor0  libxrender1
  fontconfig-config         libcolord2               libheif-plugin-aomenc libwayland-egl1     libxres1
  fonts-dejavu-core         libcups2t64              libheif1              libx11-xcb1         libxshmfence1
  fonts-dejavu-extra        libdatrie1               libhwy1t64            libxaw7             libxt6t64
```

```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ java -version
openjdk version "21.0.12.1" 2026-08-18
OpenJDK Runtime Environment (build 21.0.12.1+1-1-26.04.4-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12.1+1-1-26.04.4-Ubuntu, mixed mode, sharing)
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Add the official Jenkins repository

```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo mkdir -p /etc/apt/keyrings
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
--2026-10-12 18:49:01--  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 146.75.30.133, 2a04:4e42:95::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)|146.75.30.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1680 (1.6K) [application/octet-stream]
Saving to: '/etc/apt/keyrings/jenkins-keyring.asc'

/etc/apt/keyrings/jenkins-keyrin 100%[=========================================================>]   1.64K  --.-KB/s    in 0s

2026-10-12 18:49:01 (26.5 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc' saved [1680/1680]

(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] https://pkg
.jenkins.io/debian-stable binary/" \
| sudo tee /etc/apt/sources.list.d/jenkins.list
deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:4 https://pkg.jenkins.io/debian-stable binary/ InRelease
Get:5 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]
Get:6 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Get:7 https://pkg.jenkins.io/debian-stable binary/ Packages [31.8 kB]
Hit:8 http://security.ubuntu.com/ubuntu resolute-security InRelease
```

Install Jenkins

```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo apt install -y jenkins
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Installing:
  jenkins

Installing dependencies:
  net-tools

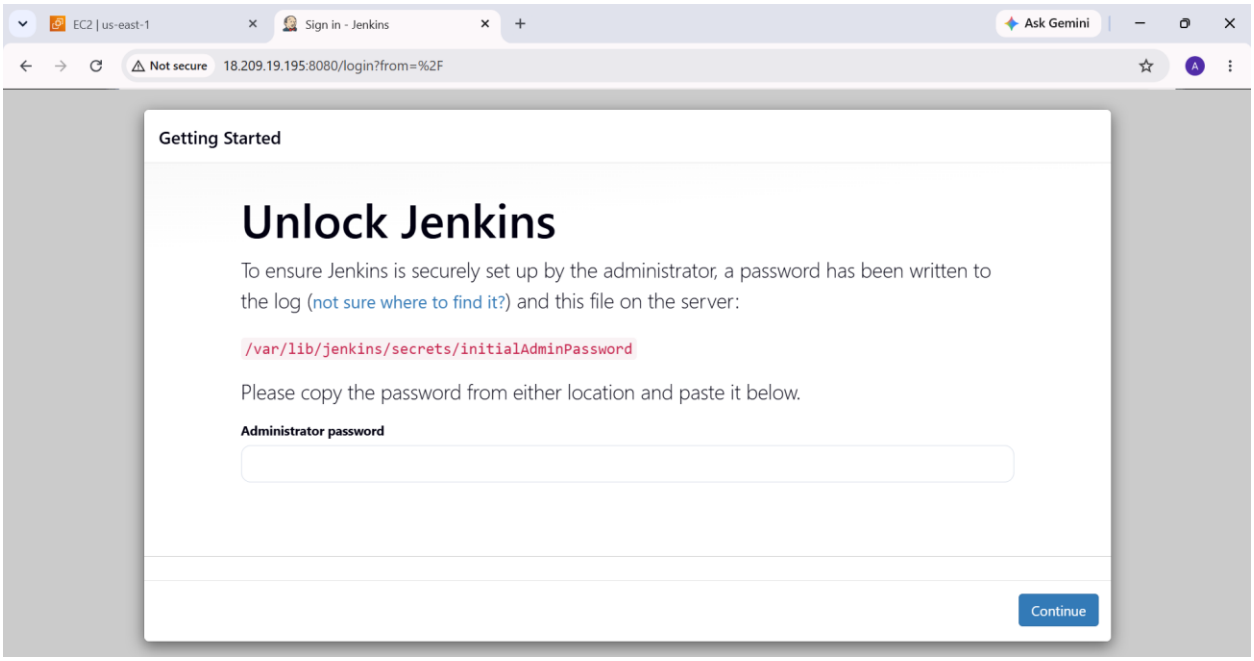
Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 2
  Download size: 53.9 MB
  Space needed: 54.7 MB / 25.7 GB available

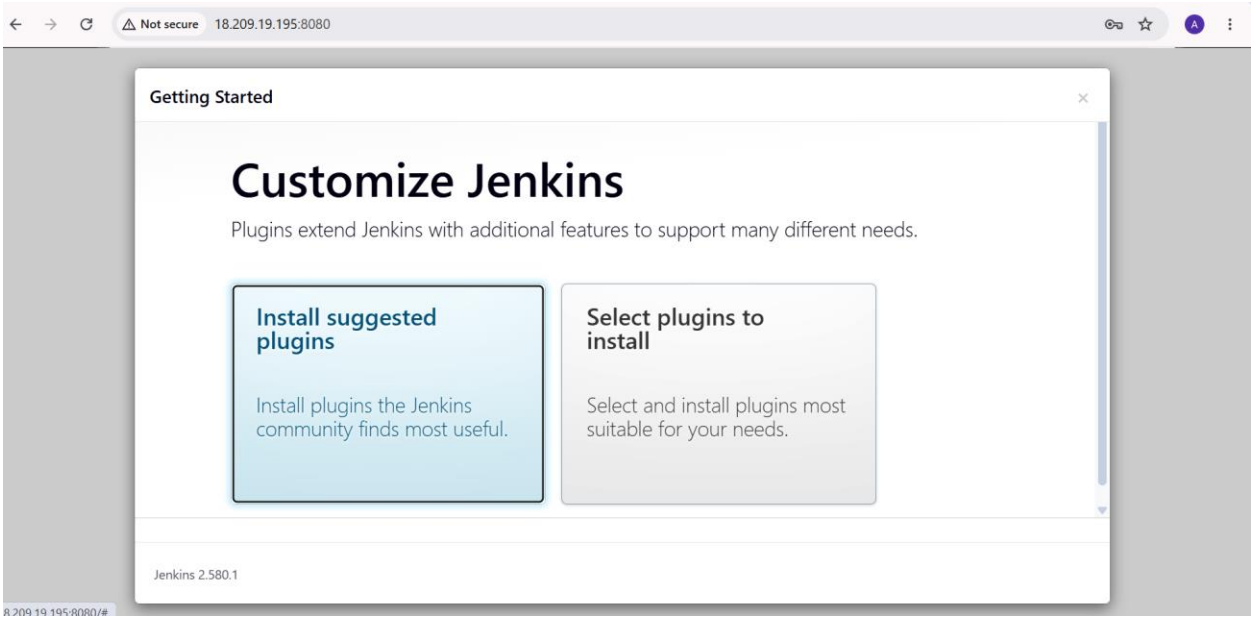
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 net-tools amd64 2.10-2ubuntu1 [152 kB]
Get:2 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.580.1 [53.7 MB]
Fetched 53.9 MB in 1s (92.4 MB/s)
Selecting previously unselected package net-tools.
(Reading database ... 136497 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-2ubuntu1_amd64v3.deb ...
Unpacking net-tools (2.10-2ubuntu1) ...
Selecting previously unselected package jenkins.
Preparing to unpack .../jenkins_2.580.1_all.deb ...
Unpacking jenkins (2.580.1) ...
Setting up net-tools (2.10-2ubuntu1) ...
```

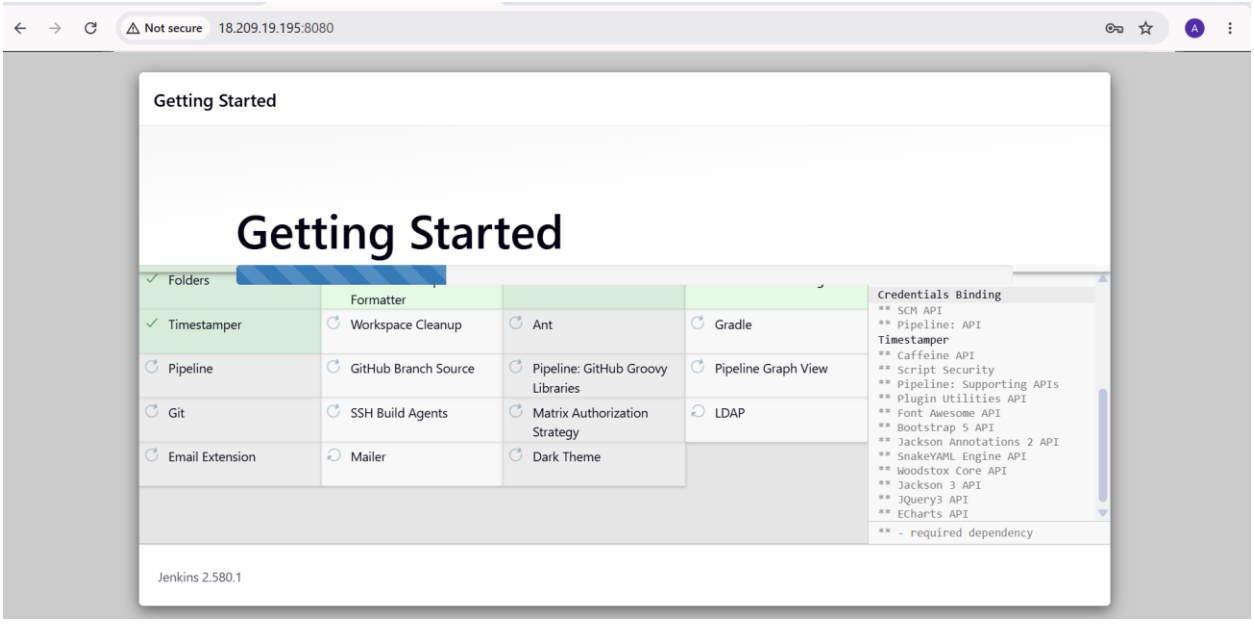
Enable Jenkins:

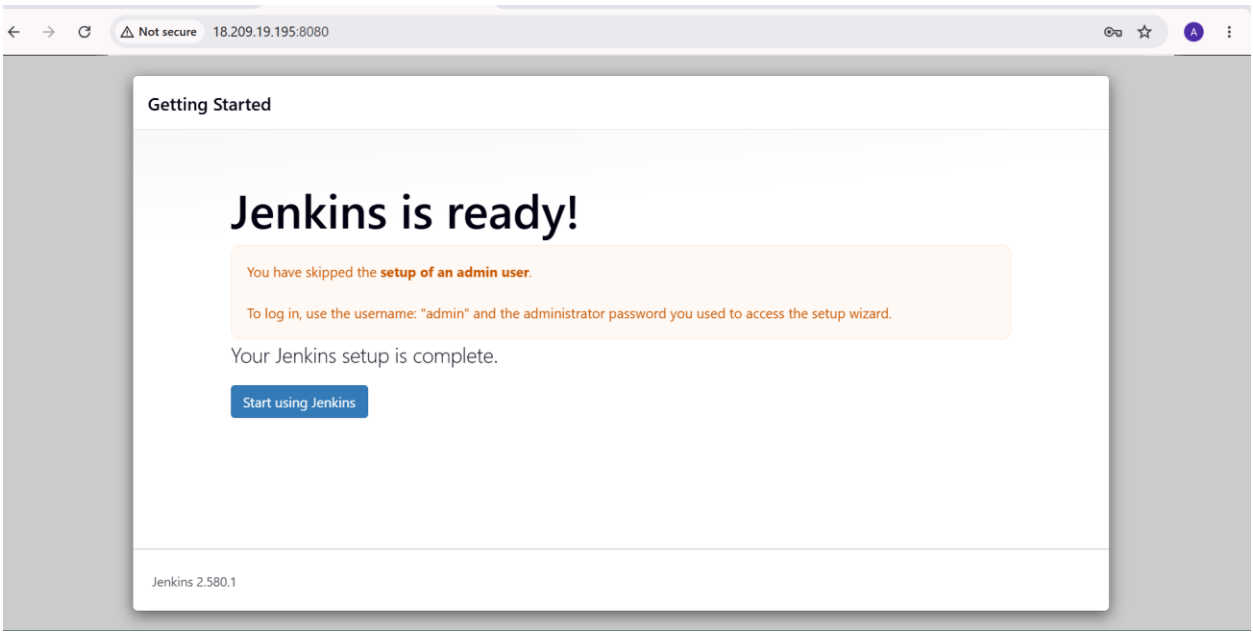
```
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo systemctl enable jenkins
sudo systemctl start jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
(.venv) ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo systemctl status jenkins --no-pager
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Mon 2026-10-12 18:50:27 UTC; 1min 59s ago
 Invocation: 800de55228b249cbaf9b1ca45e3701a6
   Main PID: 65872 (java)
      Tasks: 43 (limit: 8157)
     Memory: 599.8M (peak: 608.6M)
        CPU: 15.081s
     CGroup: /system.slice/jenkins.service
             └─65872 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --htt…
```

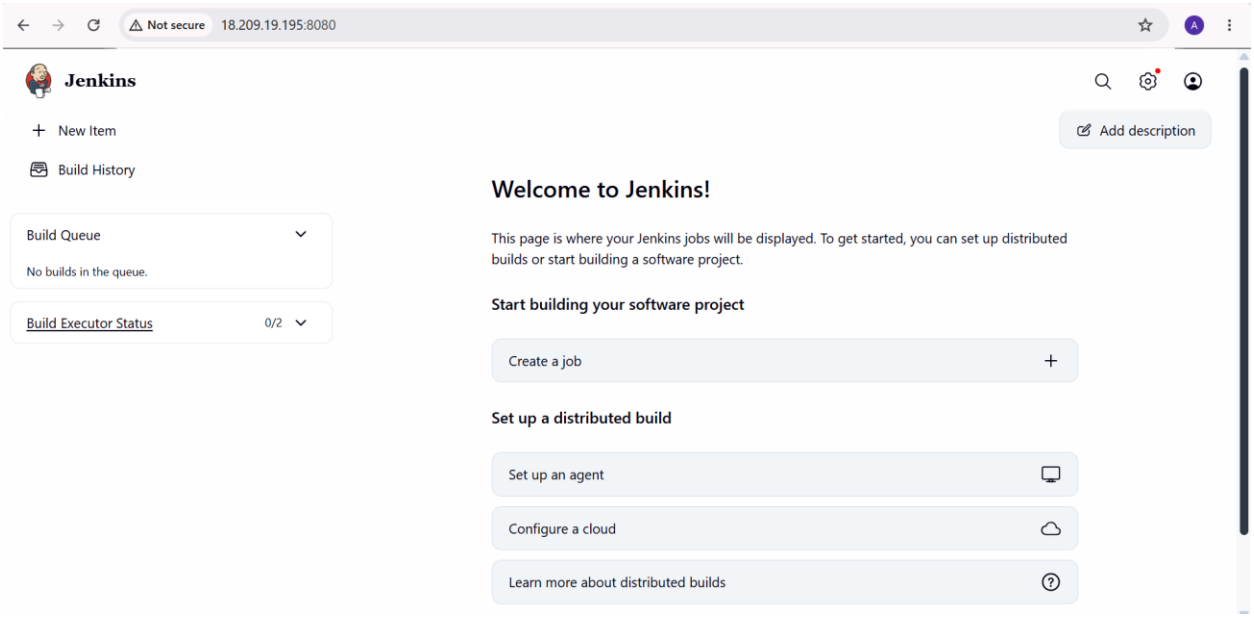
Open Jenkins and Complete Jenkins setup

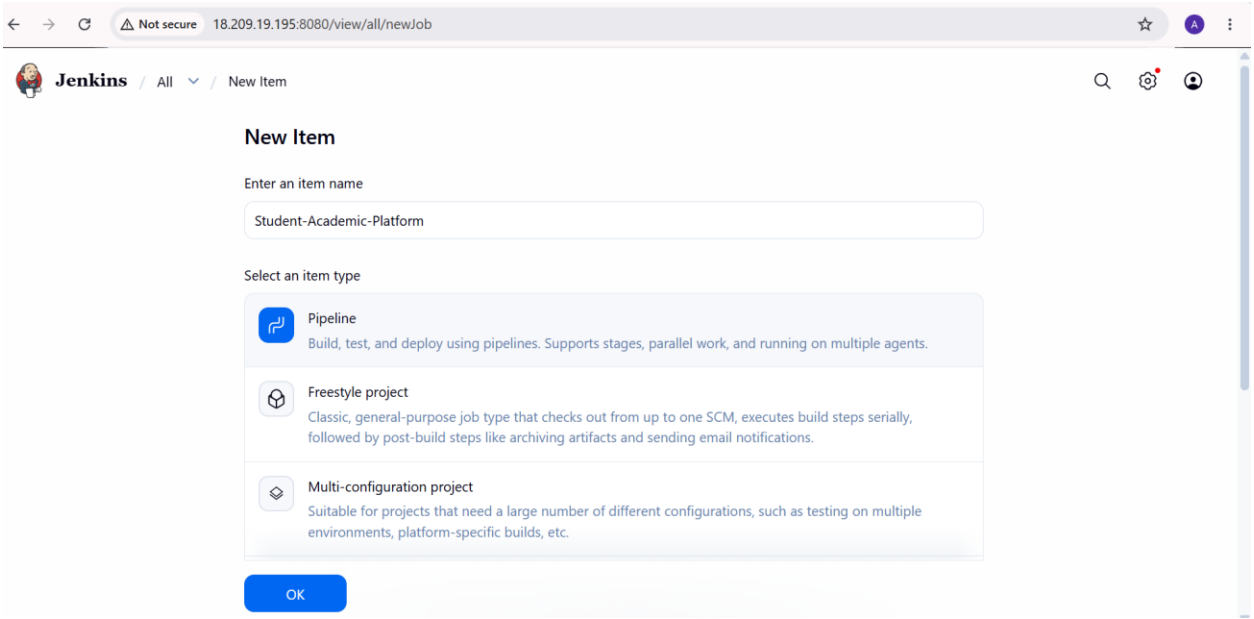


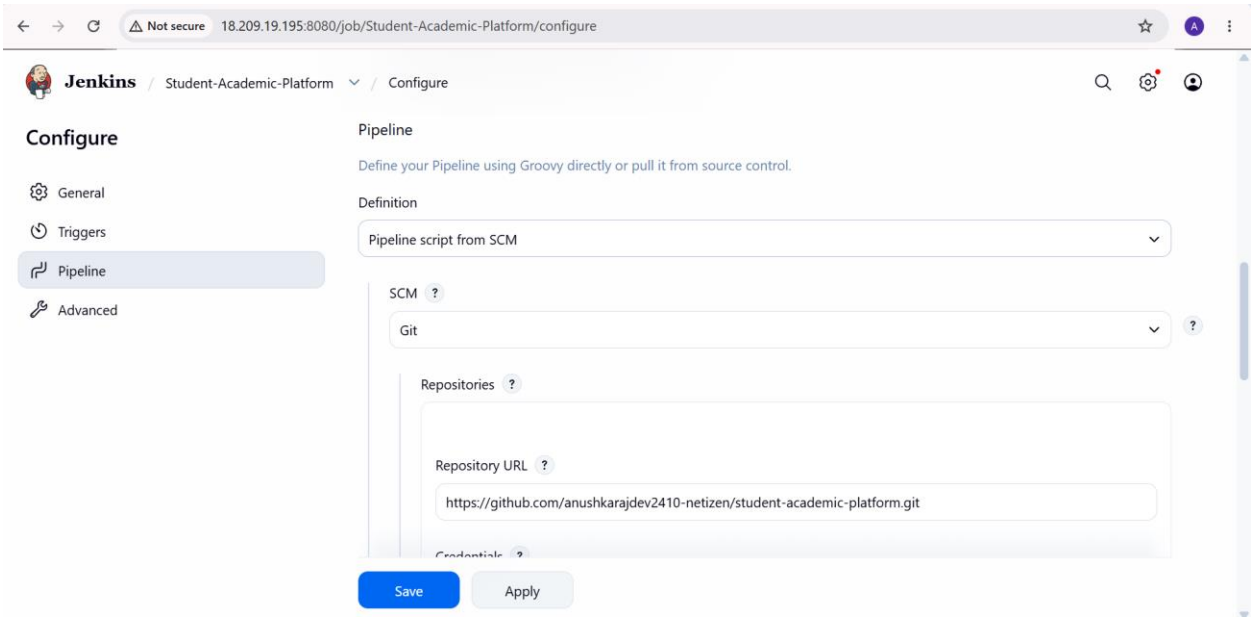

Not secure
18.209.19.195:8080
Getting Started
Customize Jenkins
Plugins extend Jenkins with additional features to support many different needs.
Install suggested plugins
Install plugins the Jenkins community finds most useful.
Select plugins to install
Select and install plugins most suitable for your needs.
Jenkins 2.580.1


Not secure
18.209.19.195:8080
Getting Started
Getting Started
Folders
Formatter
Timestamper
Workspace Cleanup
Ant
Gradle
Pipeline
GitHub Branch Source
Pipeline: GitHub Groovy Libraries
Pipeline Graph View
Git
SSH Build Agents
Matrix Authorization Strategy
LDAP
Email Extension
Mailer
Dark Theme
Credentials Binding
** SCM API
** Pipeline: API
Timestamper
** Caffeine API
** Script Security
** Pipeline: Supporting APIs
** Plugin Utilities API
** Font Awesome API
** Bootstrap 5 API
** Jackson Annotations 2 API
** SnakeYAML Engine API
** Woodstox Core API
** Jackson 3 API
** JQuery3 API
** ECharts API
** - required dependency
Jenkins 2.580.1


Not secure
18.209.19.195:8080
Getting Started
Jenkins is ready!
You have skipped the setup of an admin user.
To log in, use the username: "admin" and the administrator password you used to access the setup wizard.
Your Jenkins setup is complete.
Start using Jenkins
Jenkins 2.580.1


Not secure
18.209.19.195:8080
Jenkins
New Item
Build History
Build Queue
No builds in the queue.
Build Executor Status
0/2
Add description
Welcome to Jenkins!
This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.
Start building your software project
Create a job
Set up a distributed build
Set up an agent
Configure a cloud
Learn more about distributed builds





Jenkinsfile

```
pipeline {
    agent any

    stages {
        stage('Checkout') {
            steps {
                checkout scm
            }
        }

        stage('Install Dependencies') {
            steps {
                sh '''
                    python3 -m venv .venv
                    .venv/bin/pip install --upgrade pip
                    .venv/bin/pip install -r requirements.txt
                    .venv/bin/pip install pytest pytest-cov ruff
                '''
            }
        }

        stage('Lint') {
            steps {
                sh '.venv/bin/ruff check app tests run.py --output-format concise'
            }
```

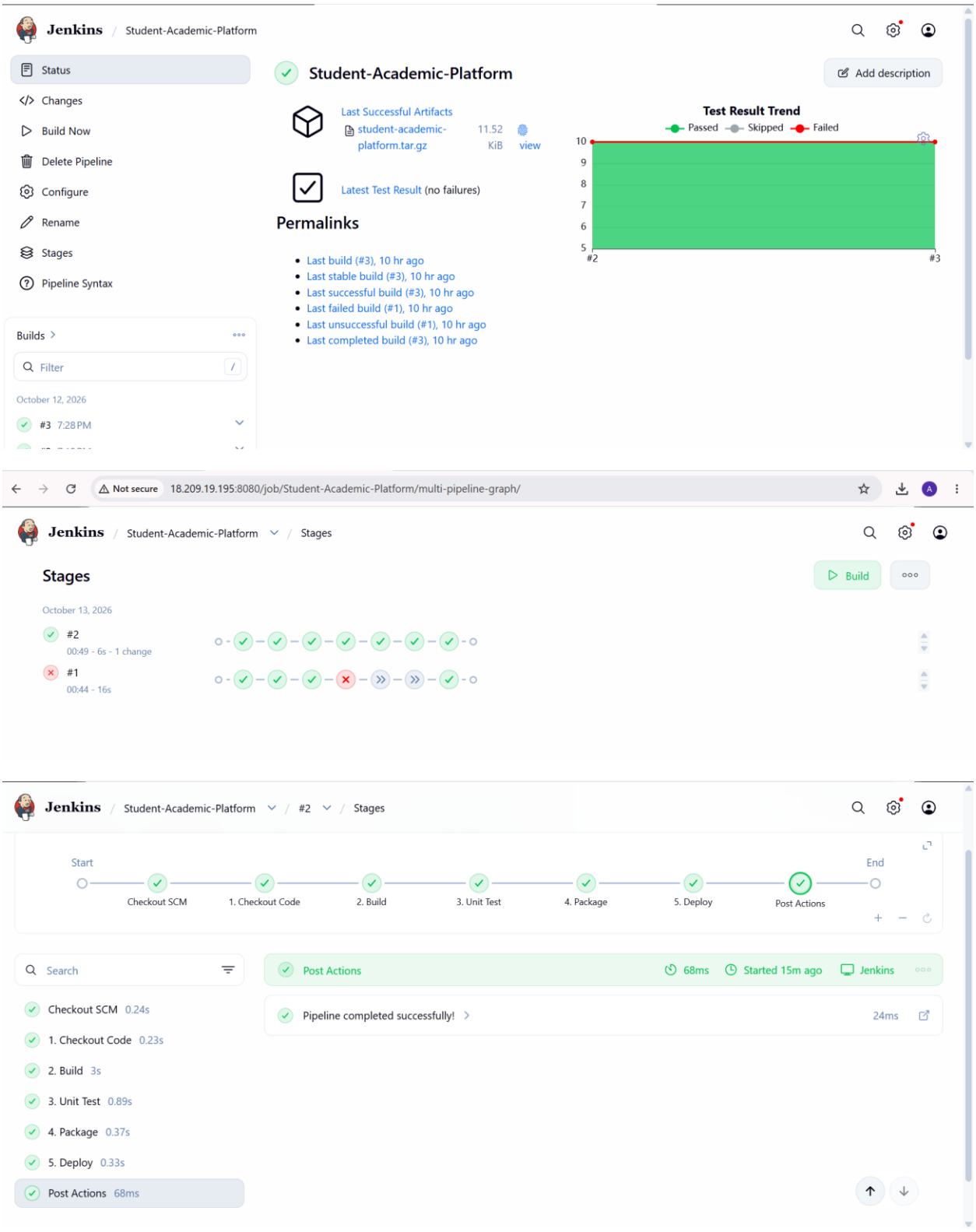
```
                steps {
                    sh '.venv/bin/ruff check app tests run.py --output-format concise'
                }
            }

            stage('Unit Tests') {
                steps {
                    sh '''
                        export PYTHONPATH="$WORKSPACE"
                        .venv/bin/python -m pytest tests/ -v \
                            --junitxml=pytest-report.xml \
                            --cov=app --cov-report=term
                    '''
                }
                post {
                    always {
                        junit allowEmptyResults: true,
                              testResults: 'pytest-report.xml'
                    }
                }
            }

            stage('Build Docker Image') {
                steps {

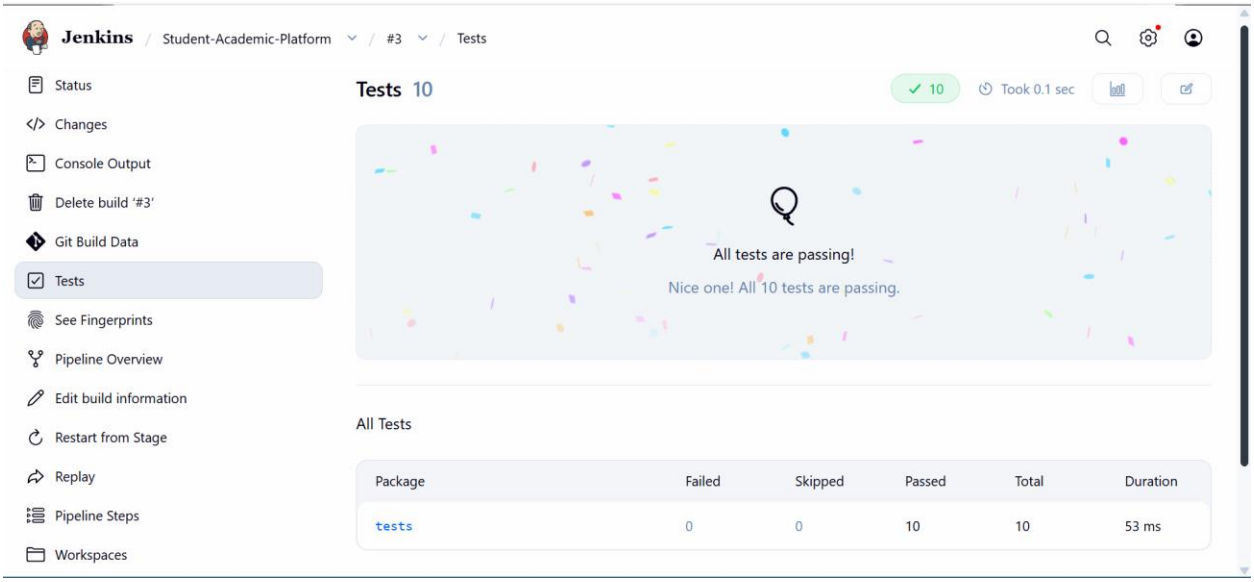
                 steps {
                     sh 'docker build -t student-academic-platform:v1.0.0 .'
                 }
             }
        }

        post {
            success {
                echo 'Tests passed and Docker image built successfully.'
            }
            failure {
                echo 'Pipeline failed. Check Console Output for details.'
            }
        }
    }
```



Download the packaged artifact

Verify the unit-test report



Deployment through Kubernetes:

Inspect the Dockerfile

```
ubuntu@ip-172-31-38-155:~/student-academic-platform$ cat Dockerfile
FROM python:3.12-slim
ENV PYTHONDONTWRITEBYTECODE=1
ENV PYTHONUNBUFFERED=1
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app ./app
COPY run.py .
RUN mkdir -p /app/database
ENV DATABASE=/app/database/student_platform.db
EXPOSE 5000
CMD ["python", "run.py"]
ubuntu@ip-172-31-38-155:~/student-academic-platform$ cat requirements.txt
Flask==3.1.1
pytest==8.4.1
pytest-cov==6.2.1
ruff==0.12.3
ubuntu@ip-172-31-38-155:~/student-academic-platform$
```

Build the Docker image

```
ubuntu@ip-172-31-38-155:~/student-academic-platform$ sudo docker build -t student-academic-platform:latest .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  91.14kB
Step 1/12 : FROM python:3.12-slim
3.12-slim: Pulling from library/python
75b6a36c64a1: Pulling fs layer
bbc5a1223747: Pulling fs layer
ecc510c1e359: Pulling fs layer
6912e23eb445: Pulling fs layer
611ca50ea732: Download complete
bbc5a1223747: Download complete
75b6a36c64a1: Download complete
97c6e8c8dabe: Download complete
6912e23eb445: Download complete
ecc510c1e359: Download complete
ecc510c1e359: Pull complete
75b6a36c64a1: Pull complete
bbc5a1223747: Pull complete
6912e23eb445: Pull complete
```

Start the application container

```
ubuntu@ip-172-31-38-155:~/student-academic-platform$ sudo docker images
                                                                                                    i Info →  U In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
python:3.12-slim                     a6e34c598f24        179MB         45.4MB
student-academic-platform:latest     6d9e541042cd        250MB         61.6MB
ubuntu@ip-172-31-38-155:~/student-academic-platform$ sudo docker run -d --name student-academic-platform --restart unless-stopped -p
 80:5000 student-academic-platform:latest
e74c19ed4389010321cdcd48cfeef7120020d00b5fde71f0c80142a3e1f2466f
ubuntu@ip-172-31-38-155:~/student-academic-platform$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND           CREATED         STATUS         PORTS
          NAMES
e74c19ed4389   student-academic-platform:latest   "python run.py"   9 seconds ago   Up 9 seconds   0.0.0.0:80->5000/tcp, [::]:80->50
00/tcp   student-academic-platform
```

Open your website

```
ubuntu@ip-172-31-38-155:~/student-academic-platform$ curl -I http://localhost
HTTP/1.1 200 OK
Server: Werkzeug/3.1.9 Python/3.12.15
Date: Mon, 12 Oct 2026 20:20:10 GMT
Content-Type: text/html; charset=utf-8
Content-Length: 25295
Connection: close
```

Complete Docker and Kubernetes/Minikube

```
ubuntu@ip-172-31-24-232:~$ curl -i http://localhost/api/health
HTTP/1.1 200 OK
Server: nginx/1.28.3 (Ubuntu)
Date: Tue, 13 Oct 2026 06:03:30 GMT
Content-Type: application/json
Content-Length: 59
Connection: keep-alive

{"service":"student-academic-platform","status":"healthy"}
ubuntu@ip-172-31-24-232:~$ cd ~/student-academic-platform
ls -la k8s
total 16
drwxrwxr-x  2 ubuntu docker 4096 Oct 12 15:25 .
drwxrwxr-x 11 ubuntu docker 4096 Oct 12 19:27 ..
-rw-rw-r--  1 ubuntu docker 1198 Oct 12 15:25 deployment.yaml
-rw-rw-r--  1 ubuntu docker  278 Oct 12 15:25 service.yaml
ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Install kubectl

```
ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Ign:5 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:6 https://pkg.jenkins.io/debian-stable binary/ Release
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [736 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [324 kB]
Fetched 1334 kB in 0s (5144 kB/s)
Reading package lists... Done
ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo apt-get install -y curl ca-certificates
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (8.18.0-1ubuntu2.7).
ca-certificates is already the newest version (20260601~26.04.1).
Solving dependencies... Done
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
ubuntu@ip-172-31-24-232:~/student-academic-platform$ KUBECTL_VERSION=$(curl -L -s https://dl.k8s.io/release/stable.txt)
ubuntu@ip-172-31-24-232:~/student-academic-platform$ curl -LO "https://dl.k8s.io/release/${KUBECTL_VERSION}/bin/linux/amd64/kubectl"
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
100 59.01M 100 59.01M   0      0 382.7M      0                              0
ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Verify:

```
ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo install -o root -g root -m 0755 kubectl /usr/local/bin/kubectl
ubuntu@ip-172-31-24-232:~/student-academic-platform$ kubectl version --client
Client Version: v1.37.1
Kustomize Version: v5.8.1
ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Install Minikube:

```
ubuntu@ip-172-31-24-232:~/student-academic-platform$ curl -LO https://storage.googleapis.com/minikube/releases/latest/minikube-linux
-amd64
  % Total    % Received % Xferd  Average Speed  Time    Time    Time   Current
                                 Dload  Upload  Total   Spent   Left   Speed
100 136.0M 100 136.0M   0      0 190.4M      0                              0
ubuntu@ip-172-31-24-232:~/student-academic-platform$ sudo install minikube-linux-amd64 /usr/local/bin/minikube
ubuntu@ip-172-31-24-232:~/student-academic-platform$ minikube version
minikube version: v1.39.0
commit: 7a9f6a841470a207de8cf4bafcccee0969d8ba10
ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Start Minikube using Docker :

```
ubuntu@ip-172-31-24-232:~/student-academic-platform$ minikube start --driver=docker
* minikube v1.39.0 on Ubuntu 26.04
* Using the docker driver based on user configuration
* Using Docker driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.51 ...
* Downloading Kubernetes v1.37.0 preload ...
    > preloaded-images-k8s-v18-v1...:  347.58 MiB / 347.58 MiB  100.00% 544.94
    > gcr.io/k8s-minikube/kicbase:  507.61 MiB / 507.61 MiB  100.00% 253.96 MiB
* Creating docker container (CPUs=2, Memory=3072MB) ...
* Preparing Kubernetes v1.37.0 on containerd 2.3.4 ...
* Configuring CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, default-storageclass
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
ubuntu@ip-172-31-24-232:~/student-academic-platform$ kubectl get nodes
NAME       STATUS     ROLES           AGE   VERSION
minikube   NotReady   control-plane   12s   v1.37.0
ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

```
ubuntu@ip-172-31-24-232:~/student-academic-platform$ kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   2m28s   v1.37.0
ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Build the image inside Minikube:

```
ubuntu@ip-172-31-24-232:~/student-academic-platform$ minikube image build -t student-academic-platform:v1.0.0 .
#1 [internal] load build definition from Dockerfile
#1 transferring dockerfile: 342B done
#1 DONE 0.0s

#2 [internal] load metadata for docker.io/library/python:3.12-slim
#2 DONE 0.4s

#3 [internal] load .dockerignore
#3 transferring context: 115B done
#3 DONE 0.0s

#4 [1/7] FROM docker.io/library/python:3.12-slim@sha256:a6e34c598f2467ed0e9a8d349809fcd8b5c603269512df273a0bb1784edc11b1
#4 resolve docker.io/library/python:3.12-slim@sha256:a6e34c598f2467ed0e9a8d349809fcd8b5c603269512df273a0bb1784edc11b1 0.0s done
#4 ...

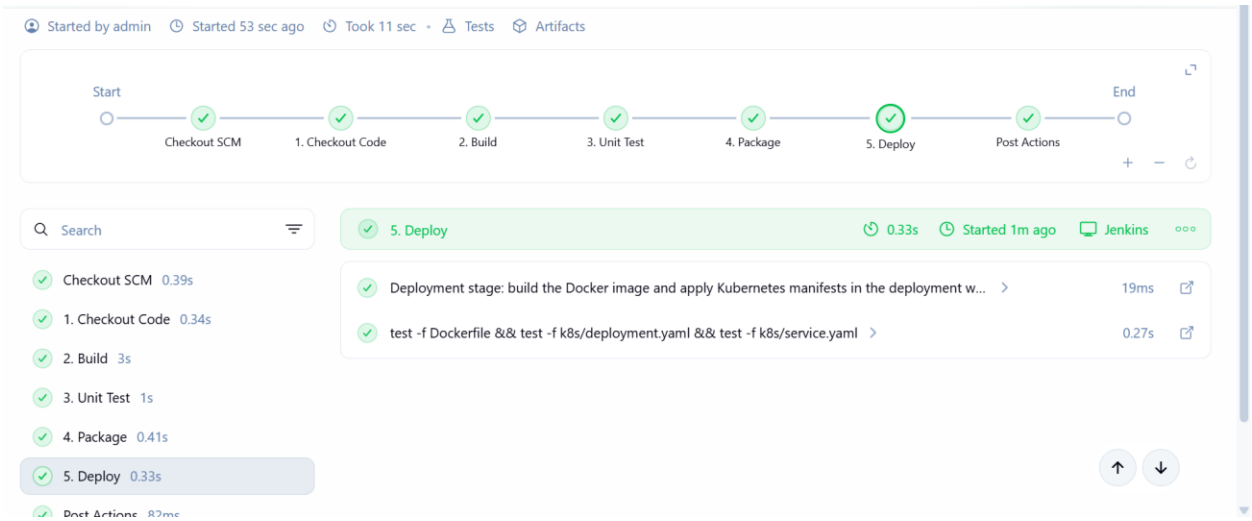
#5 [internal] load build context
#5 transferring context: 56.87kB done
#5 DONE 0.1s
```

```
ubuntu@ip-172-31-24-232:~/student-academic-platform$ minikube image ls
registry.k8s.io/pause:3.10.2
registry.k8s.io/kube-scheduler:v1.37.0
registry.k8s.io/kube-proxy:v1.37.0
registry.k8s.io/kube-controller-manager:v1.37.0
registry.k8s.io/kube-apiserver:v1.37.0
registry.k8s.io/etcd:3.7.0-0
registry.k8s.io/coredns/coredns:v1.14.6
gcr.io/k8s-minikube/storage-provisioner:v5
docker.io/library/student-academic-platform:v1.0.0
docker.io/kindest/kindnetd:v20260820-69b56db7
docker.io/kindest/kindnetd:v20250512-df8de77b
ubuntu@ip-172-31-24-232:~/student-academic-platform$
```

Deploy the application:

```
ubuntu@ip-172-31-24-232:~/student-academic-platform$ kubectl apply -f k8s/deployment.yaml
deployment.apps/student-academic-deployment created
ubuntu@ip-172-31-24-232:~/student-academic-platform$ kubectl apply -f k8s/service.yaml
service/student-academic-service created
ubuntu@ip-172-31-24-232:~/student-academic-platform$ kubectl get deployments
NAME                         READY   UP-TO-DATE   AVAILABLE   AGE
student-academic-deployment   3/3     3            3           16s
ubuntu@ip-172-31-24-232:~/student-academic-platform$ kubectl get pods -o wide
NAME                                          READY   STATUS    RESTARTS   AGE   IP           NODE       NOMINATED NODE   READINESS
 GATES
student-academic-deployment-68dfdffcbd-44pgp   1/1     Running   0          24s   10.244.0.4   minikube   <none>           <none>
student-academic-deployment-68dfdffcbd-4lqjb   1/1     Running   0          24s   10.244.0.5   minikube   <none>           <none>
student-academic-deployment-68dfdffcbd-6zhv6   1/1     Running   0          24s   10.244.0.3   minikube   <none>           <none>
ubuntu@ip-172-31-24-232:~/student-academic-platform$ kubectl get services
NAME                     TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
kubernetes               ClusterIP   10.96.0.1       <none>        443/TCP          8m4s
student-academic-service   NodePort    10.111.171.97   <none>        5000:30090/TCP   23s
```

```
ubuntu@ip-172-31-24-232:~/student-academic-platform$ kubectl rollout status deployment/student-academic-deployment --timeout=120s
deployment "student-academic-deployment" successfully rolled out
ubuntu@ip-172-31-24-232:~/student-academic-platform$
```


Started by admin
Started 53 sec ago
Took 11 sec
Tests
Artifacts
Start
Checkout SCM
1. Checkout Code
2. Build
3. Unit Test
4. Package
5. Deploy
Post Actions
End
Search
Checkout SCM 0.39s
1. Checkout Code 0.34s
2. Build 3s
3. Unit Test 1s
4. Package 0.41s
5. Deploy 0.33s
5. Deploy
0.33s
Started 1m ago
Jenkins
Deployment stage: build the Docker image and apply Kubernetes manifests in the deployment w...
19ms
test -f Dockerfile && test -f k8s/deployment.yaml && test -f k8s/service.yaml
0.27s